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Lecture 11: Classification

ML/AI for Engineering Applications
ENV 6932

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Fall 2025

Classification

- Recall
- Classification Metrics

Two common curves as classification metrics

- ROC: Receiver Operating Characteristic
- AUC: Area Under Curve

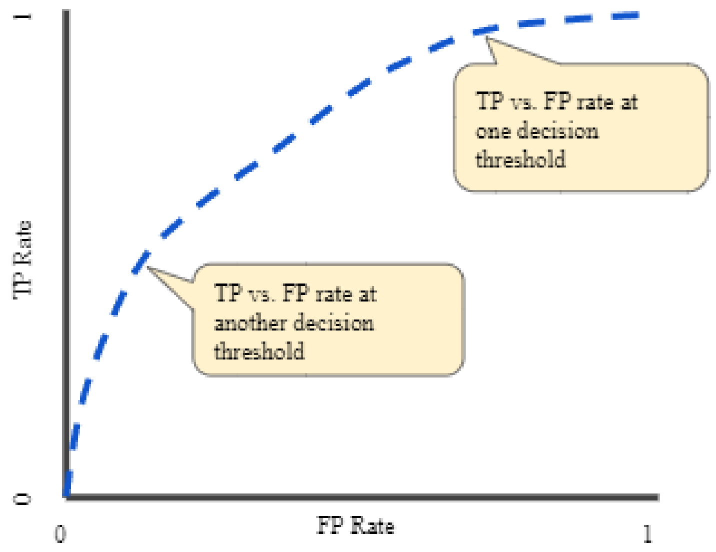
ROC curve: Receiver Operating Characteristic

- Roc: a graph showing the performance of a classification model at all classification thresholds.
- True Positive Rate
- False Positive Rate

$$TPR = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{FP + TN}$$

		Predicted Class	
		NO	YES
Actual Class	NO	True Negative (TN)	False Positive (FP)
	YES	False Negative (FN)	True Positive (TP)



https://towardsdatascience.com/demystifying-roc-curves-df809474529a
https://developers.google.com/machine-learning/crash-course/classification/roc-and-auc

ROC curve: Receiver Operating Characteristic

- ROC is an important assistant in evaluating and fine-tuning classification models.
- A bit hard to understand for beginners.
- It shows the performance of one classification model at all classification thresholds.
- Application:
  - can be used to evaluate the strength of a model.
  - to compare two classification models.

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ROC: The name origin

- First used during World War II for the analysis of radar signals after the attack on Pearl Harbor.
- The US army wanted to improve the rate of detection of Japanese aircraft from their radar signals.
- They measured the ability of a radar receiver operator to make these predictions called the Receiver Operating Characteristic.
- The goal was to analyze the predictive power of the predictor in confirming the detection of as many true positives as possible while minimizing false positives.

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Why comparing TPR and FPR?

- Example: consider a binary classification that predicts probability (Pr) of observations (obs) belonging to Class YES and NO.
- Choose a threshold = 0.5. Observations with Pr >= 0.5 belong to Class Yes and Pr <0.5 belong to Class NO.

$$TPR = \frac{True\ Positive}{True\ Positive + False\ Negative}$$

$$FPR = \frac{False\ Positive}{False\ Positive + True\ Negative}$$

		Predicted Class	
		NO	YES
Actual Class	NO	True Negative (TN)	False Positive (FP)
	YES	False Negative (FN)	True Positive (TP)

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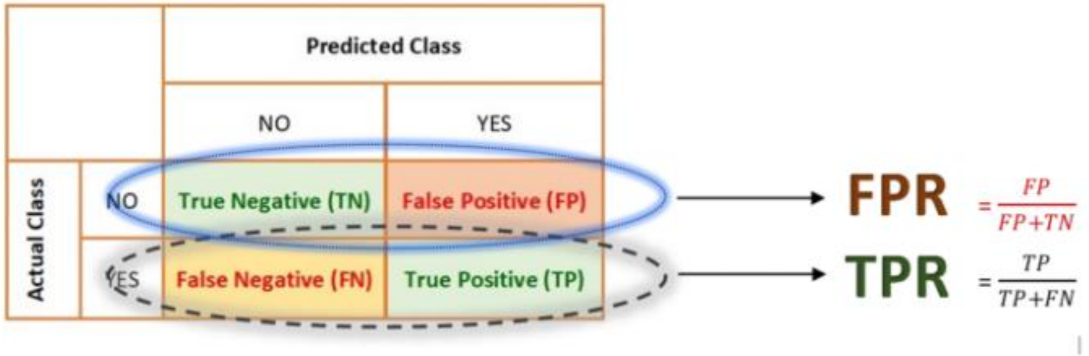
Why comparing TPR and FPR?

- TPR and FPR are derived from two separate rows of the confusion matrix.
- The True Positive Rate looks at the **actual YESs** and the False Positive Rate looks at the **actual NOs**.



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Purpose of ROC Curve

- **Purpose 1 - Analyzing the strength/predictive power of a classifier**
- The task of a classifier is to assign higher Pr to observations that belong to class YES and lower Pr to observations (obs) that belong to class NO.
- Therefore, if there is a considerable differences in the probabilities assigned to obs of either class, we will be able to define a threshold and predict the two classes with high confidence.
- The ROC of a good classifier is closer to **the top left of the graph**.
- **WHY?** Since a good classifier achieves high TPR at low FPR (at optimal threshold).

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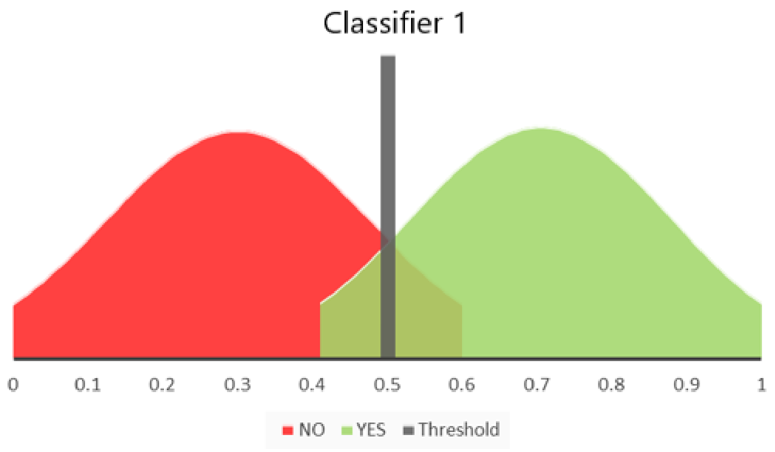
Why comparing TPR and FPR?

- The green curve shows the distribution of Class YES. The red curve: distribution of Class NO over the probability assigned to each observation by the classifier.
- If we choose threshold 0.5 we are able to classify most observations correctly. There is some overlap though. The overlapping cases will get misclassified at 0.5.

- How happens if selecting the threshold to 0.6? it classifies almost 85% true positives correctly and with 0 false positives.
- How about lowering the threshold from 0.6 to 0.5? the TPR will increase but so will the FPR.



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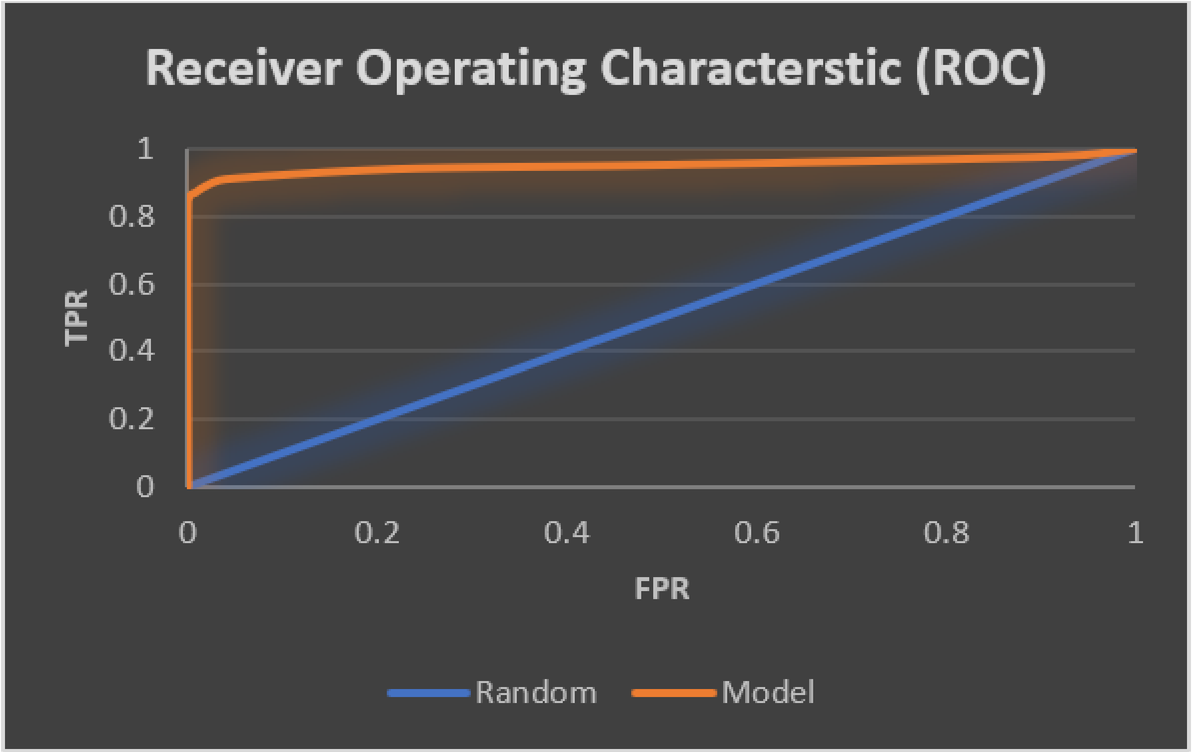
Purpose of ROC Curve

- Note: in the ROC curve below the thresholds are not shown.



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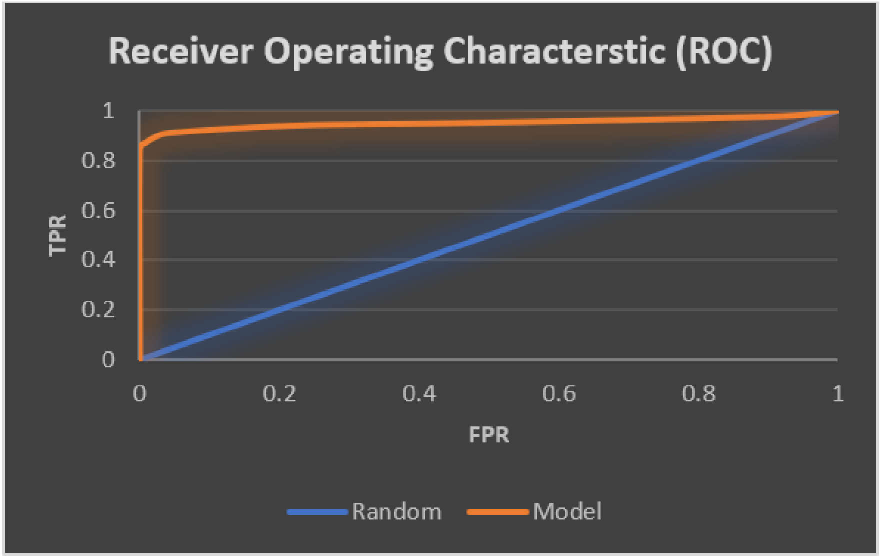
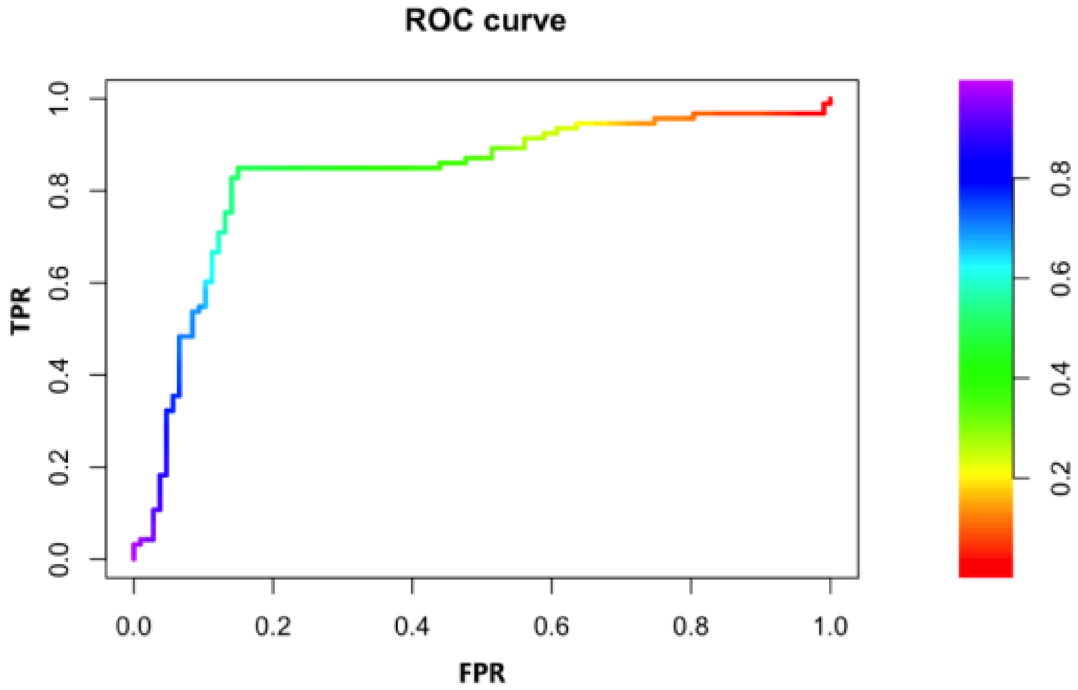
Purpose of ROC Curve

- Note: in the ROC curve below the thresholds are not shown.
- So, we can use a color-coded ROC curve.





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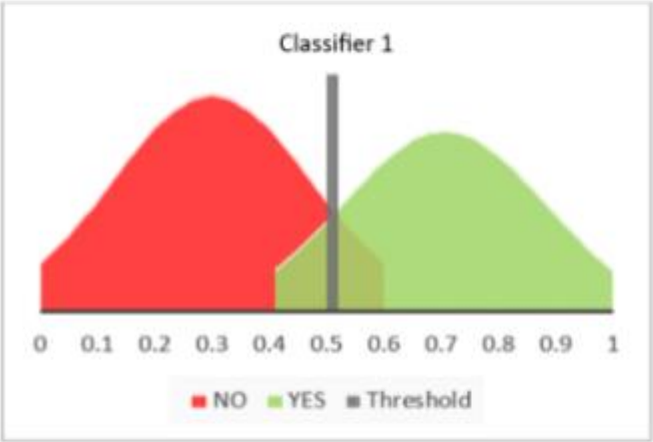
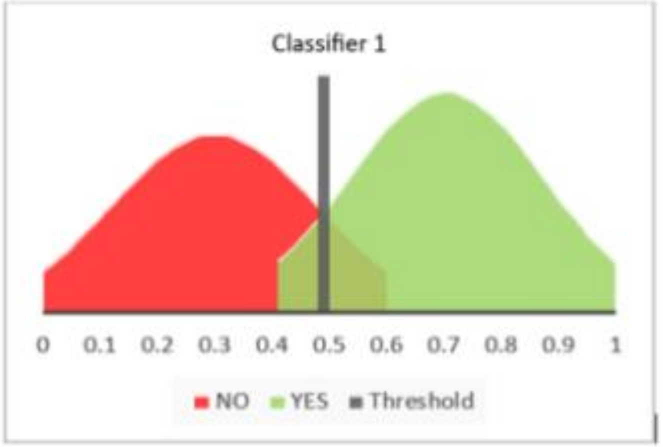
Purpose of ROC Curve

- The prevalence/class distribution does not affect ROC curves.
- Classifier #1 separates classes the same way regardless of if the dataset is in-balanced dataset or balanced dataset. The ROC Curve for this classifier will still be the same.





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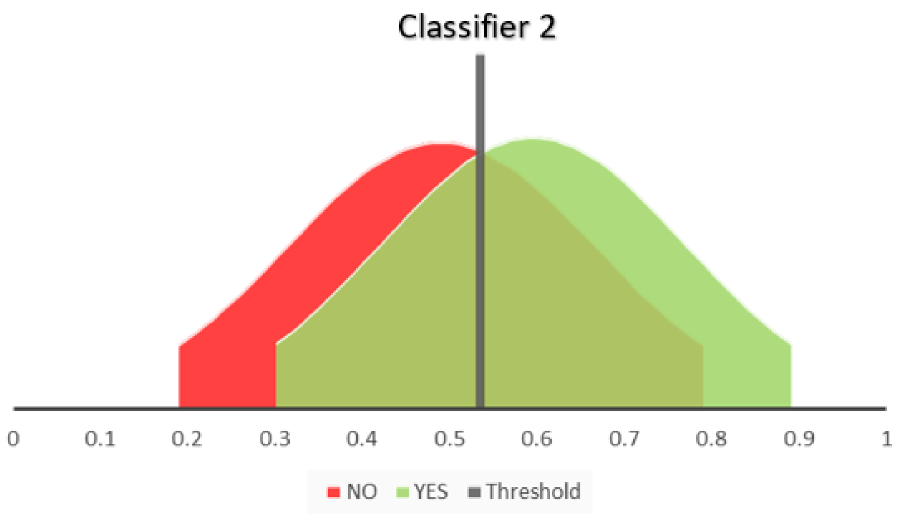
A Poor Classifier

- A poor classifier does not distinguish the two classes well.
- The ROC curve for Classifier 2 shown below will be closer to the diagonal.
- Implying lower TPR rates (more false negatives) and higher FPR rates (more false positives)



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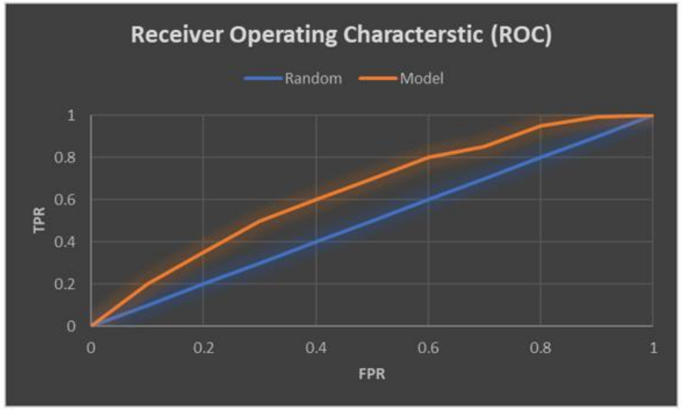

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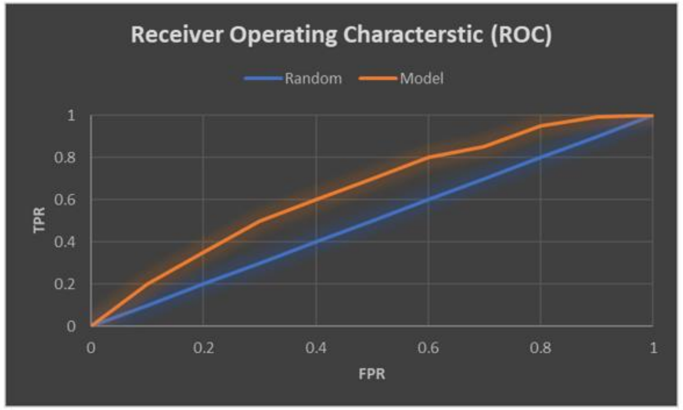
A Poor Classifier

- The Random model (shown in blue diagonal) is a random classifier that does not have any ability to distinguish between the two classes.
- The predicted probabilities of the two classes overlap (Classifier 3). TPR = FPR at any threshold.



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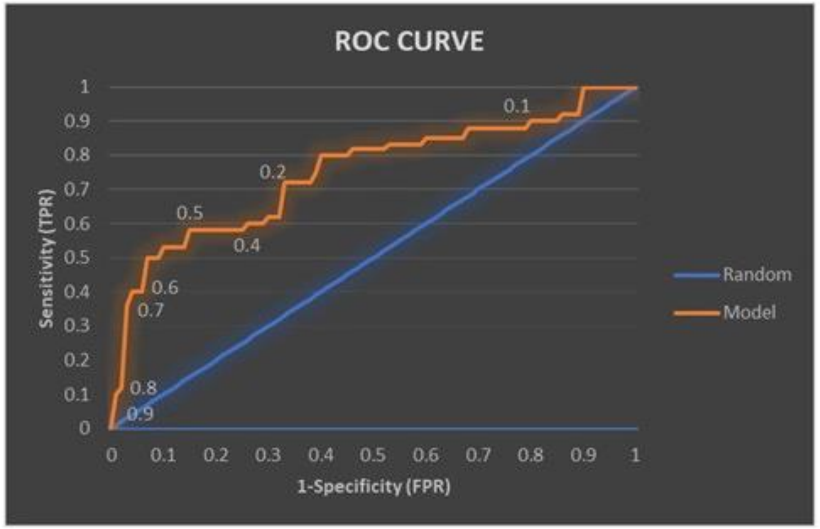
Purpose of ROC Curve

- **Purpose 2 - Determining optimal threshold**
- It helps find the threshold where the TPR is high and FPR is low i.e. misclassifications are low.
- In many cases, the cost of FP and FN is not the same. The ROC becomes extremely useful.



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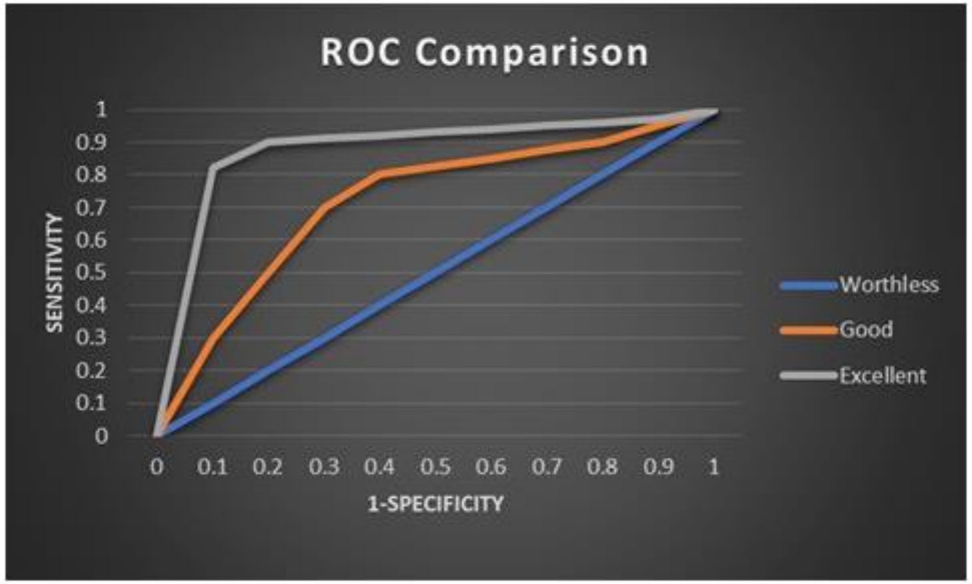
Purpose of ROC Curve

- **Purpose 3 - Comparing two models (using Area Under the Curve)**
- The closer is an ROC to the 45-degree diagonal, the less powerful is the model.



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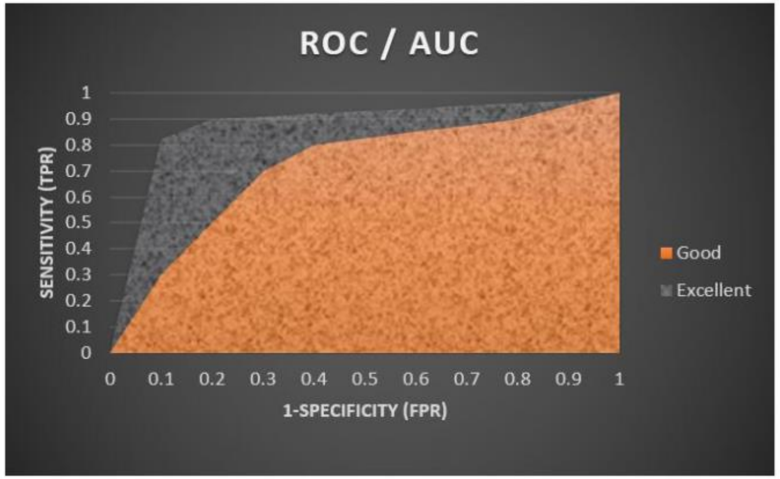
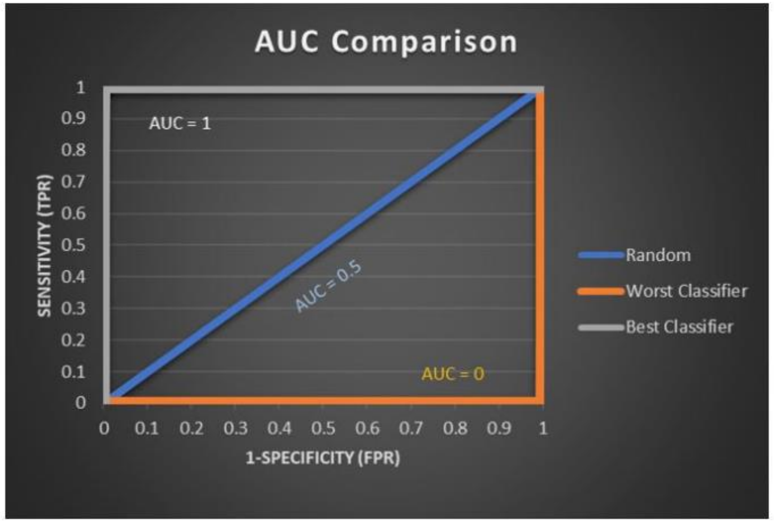
AUC - Area Under the Curve

- AUC offers an aggregate measure of performance of a model **across all possible thresholds**.
- AUC shows a measure of separability.
- It shows the capability of the model in distinguishing between classes.
- Higher the AUC, better the model is at predicting the probability of class YES higher than the probability of class NO.
- AUC ranges from 0 to 1. A model whose predictions are 100% right has an AUC of 1.0; one whose predictions are 100% wrong has an AUC of 0.0.





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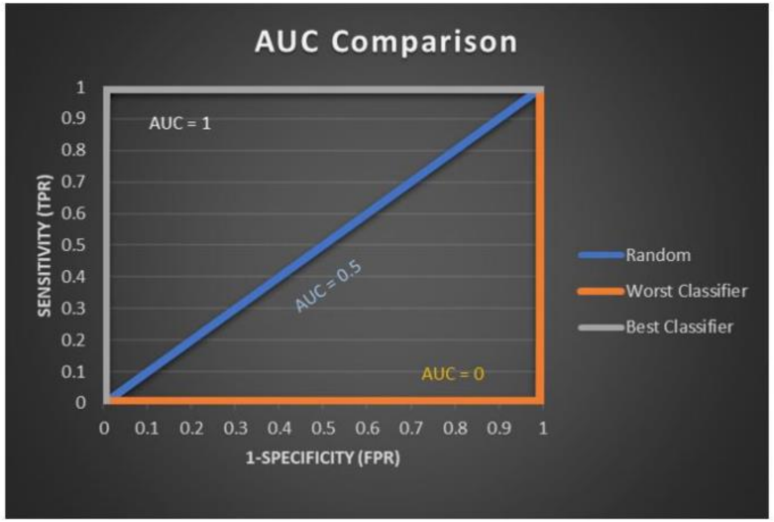
Is the random model the worst possible model?

- The random classifier predicts an observation as class YES or NO at random. The AUC would be 0.5 and TPR =FPR at all thresholds.
- The worst possible model would be a model that predict all the class YES obs as class NO and class NO obs as class YES. The AUC of such a model would be 0.



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References

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